Word Embedding: Representing meaning of the words

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How smart are you?

- **king:queen :: man:?**

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- **king:queen :: man:?**
- woman

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- How can a computer know this?

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How does Google know these words are related?

Represent the words so that computers can understand the meaning

Outline

- Semantics = Meaning
- Vector Semantics
  - Word vectors
  - tf-idf
  - Word Embeddings
    - Word2vec
    - BERT

Ref: Chapter-6, SLP3, Jurafsky
(https://web.stanford.edu/~jurafsky/slp3/6.pdf)

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- Giving a complete definition to a simple entity is difficult. Then what about for machines!

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- **Computational semantics** is the study of how computers represent, process, and understand the meaning of words, phrases, and sentences using computational methods.

Motivation

Suppose you didn't know the meaning of the word **ongchoi** but you see it in the following contexts:

- *Ongchoi* `is delicious sauteed with garlic.`
- *Ongchoi* `is superb over rice.`
- `...ongchoi leaves with salty sauces...`

And suppose that you had seen many of these context words in other contexts:

- `...spinach sauteed with garlic over rice...`
- `...chard stems and leaves are delicious...`
- `...collard greens and other salty leafy greens`
- The fact that ongchoi occurs with words like rice and garlic and delicious and salty, as do words like spinach, chard, and collard greens might suggest that *ongchoi* is a leafy green similar to these other leafy greens
- We can computationally do the same by using the words neighboring *ongchoi*

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  `I saw a dog` **`run`** `across the field.`
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  `I drive a vehicle to work every day.`

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  `I drive a car to work every day.`
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- Semantically similar words have similar distribution pattern

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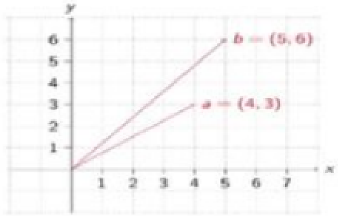
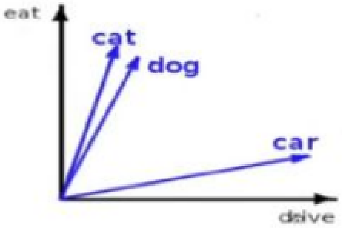
- Distributions are vectors in a multidimensional semantic space





Example

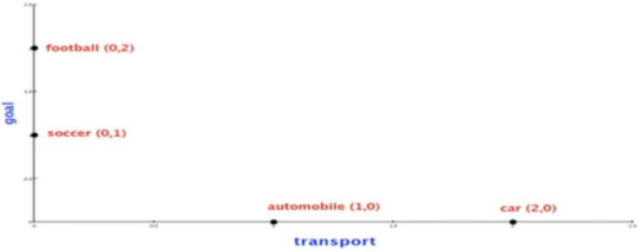
Small Dataset

An automobile is a wheeled motor vehicle used for transporting passengers .
A car is a form of transport , usually with four wheels and the capacity to carry around five passengers .
Transport for the London games is limited , with spectators strongly advised to avoid the use of cars .
The London 2012 soccer tournament began yesterday , with plenty of goals in the opening matches .
Giggs scored the first goal of the football tournament at Wembley , North London .
Bellamy was largely a passenger in the football match , playing no part in either goal .

Target words: ⟨automobile, car, soccer, football⟩
Term vocabulary: ⟨wheel, transport, passenger, tournament, London, goal, match⟩

distributional matrix = targets X contexts

	wheel	transport	passenger	tournament	London	goal	match
automobile	1	1	1	0	0	0	0
car	1	2	1	0	1	0	0
soccer	0	0	0	1	1	1	1
football	0	0	1	1	1	2	1



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Using simple vector product

automobile . car = 4
automobile . soccer = 0
automobile . football = 1

car . soccer = 1
car . football = 2
soccer . football = 5

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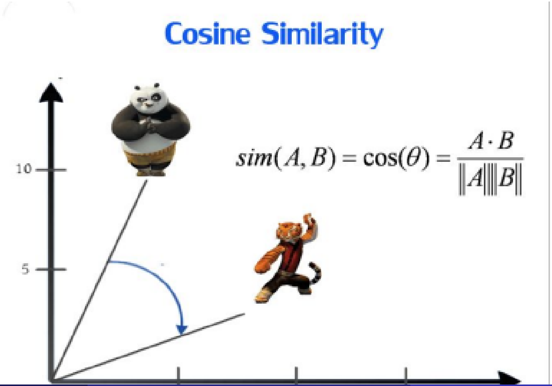
- Objective: words with similar context occupy close spatial positions



One-hot embedding

Have a good day and Have a great day. They hardly have different meaning.
If we construct an exhaustive vocabulary (let's call it V), it would have V $= \{$Have, a,good, great, day$\}$.
One-hot encoding

Have = [1,0,0,0,0]\` ; a=[0,1,0,0,0]\` ; good=[0,0,1,0,0]\` ; great= [0,0,0,1,0]\` ; day=[0,0,0,0,1]\` (\` represents transpose)

If we project this vectors in 5D space, we can see good and great have difference.. Which is not True

How to generate Distributional Representaion

1. Pre-process a corpus (to define targets and contexts)
2. Select targets and contexts
3. Count the target-context co-occurrences
4. Weight the contexts (optional)
5. Build the distributional matrix
6. Reduce the matrix dimensions
7. Compute the vector distances on the (reduced) matrix

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- Two approaches:
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  - Pointwise Mutual Information (PMI)
    - Measure of how often two events x and y occur, compared with what we would expect if they were independent
    - $PMI(w, c) = log_2 \frac{P(w, c)}{P(w)P(c)}$
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- Prediction Based Methods (Neural network based embedding)
  - Continuous Bag-of-Words (CBOW)
  - Skip-gram

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- Instead of counting how often each word w occurs near, say, c, we'll instead train a classifier on a binary prediction task: "Is word w likely to show up near c?"
- Use running text as implicitly supervised training data for such a classifier **(Self-supervision)**

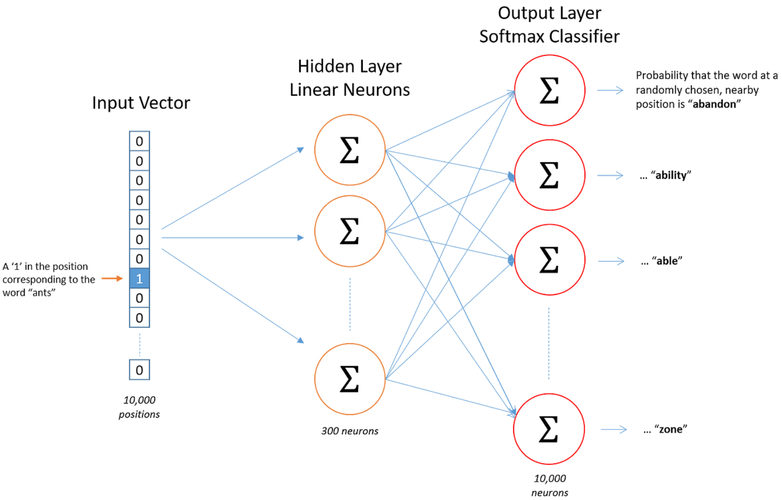
Skip-gram Method

- Tries to predict the context given a target word
- given the center word “jumped”, the model will be able to predict or generate the surrounding words “The”, “cat”, “over”, “the”, “puddle”.

Method:

- reat the target word and a neighboring context word as positive examples.
- Randomly sample other words in the lexicon to get negative samples.
- Use logistic regression to train a classifier to distinguish those two cases.
- Use the learned weights as the embeddings

Skip-gram Method



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  $$P(+|w,c) = \sigma(c \ dot \ w) = \frac{1}{1+exp(c \ dot \ w)}$$

Skip-gram Method

- There are many context words in the window. Skip-gram makes the simplifying assumption that all context words are independent

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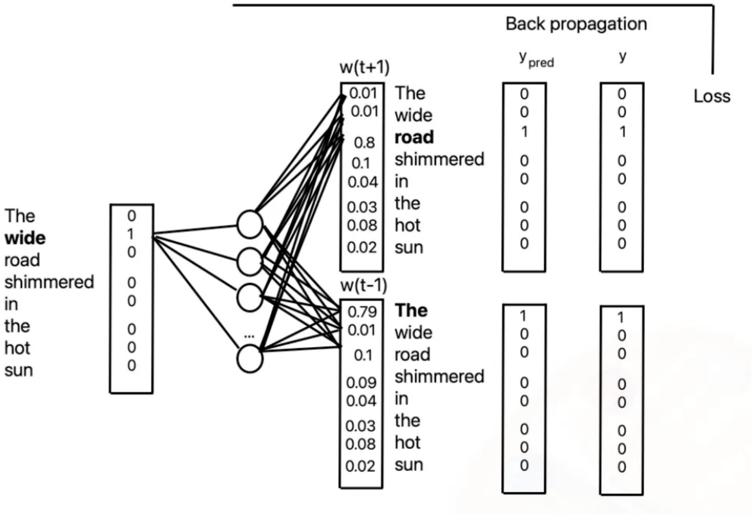
- Skip-gram actually stores two embeddings for each word, one for the word as a target, and one for the word considered as context
- Use cross entropy loss.

Skip-gram Method [1]

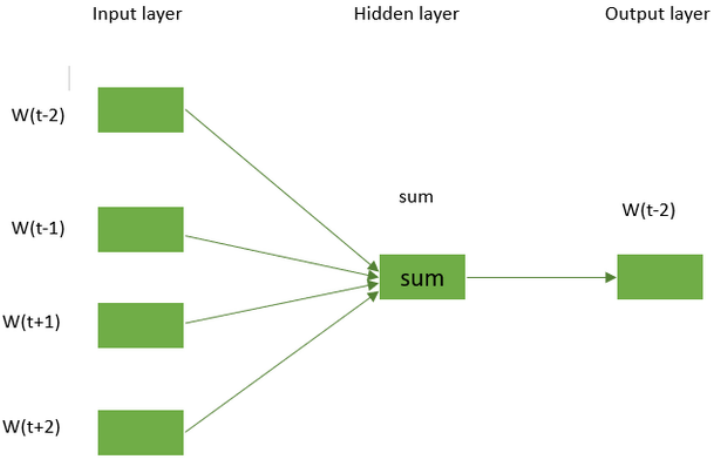


[1] https://medium.com/@corymaklin/word2vec-skip-gram-904775613b4c

CBOW Method

- Predict the probability of a word given a context ($P(w|t)$)
- Context may be single or group of words
- Lets see this by an example `The cat jumped over the puddle.`
- So one approach is to treat “The”, “cat”, 'over”, “the', “puddle” as a context and from these words, be able to predict or generate the center word “jumped”.
- This type of model we call a Continuous Bag of Words (CBOW) Model.

Skip-gram Method [2]



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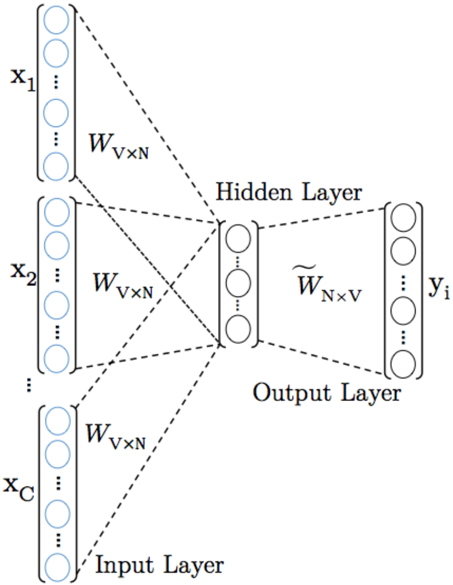
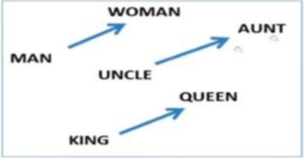


$\mathbf{x}_1$
$W_{V\times N}$
$\mathbf{x}_2$
$W_{V\times N}$
$\mathbf{x}_C$
$W_{V\times N}$
Hidden Layer
$\widetilde{W}_{N\times V}$
$\mathbf{y}_i$
Output Layer
Input Layer

Learned vectors can be used for reasoning



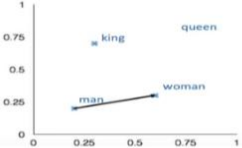
Analogy Testing

Relationship	Example 1	Example 2	Example 3
France - Paris	Italy: Rome	Japan: Tokyo	Florida: Tallahassee
big - bigger	small: larger	cold: colder	quick: quicker
Miami - Florida	Baltimore: Maryland	Dallas: Texas	Kona: Hawaii
Einstein - scientist	Messi: midfielder	Mozart: violinist	Picasso: painter
Sarkozy - France	Berlusconi: Italy	Merkel: Germany	Koizumi: Japan
copper - Cu	zinc: Zn	gold: Au	uranium: plutonium
Berlusconi - Silvio	Sarkozy: Nicolas	Putin: Medvedev	Obama: Barack
Microsoft - Windows	Google: Android	IBM: Linux	Apple: iPhone
Microsoft - Ballmer	Google: Yahoo	IBM: McNealy	Apple: Jobs
Japan - sushi	Germany: bratwurst	France: tapas	USA: pizza

a:b :: c:? $\longrightarrow$ $\boxed{d = \arg\max_x \frac{(w_b - w_a + w_c)^T w_x}{||w_b - w_a + w_c||}}$

man:woman :: king:?

+	king	[0.30 0.70]
-	man	[0.20 0.20]
+	woman	[0.60 0.30]
	queen	[0.70 0.80]



Demo- Word2vec

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- **BERT!**

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- BERT is a deep neural language model that generates contextual embeddings by considering both the left and right context of every word in a sentence.
- Internally, BERT uses a Transformer architecture. The Transformer allows every word to look at the other words in the sentence before generating its embedding. The details of Transformers are a separate topic.

Demo- BERT

Summary

- Computers understand meaning of words by understanding the similarity between the words (in a distribution)
- Word vectors/embedding helps to capture this similarity effectively